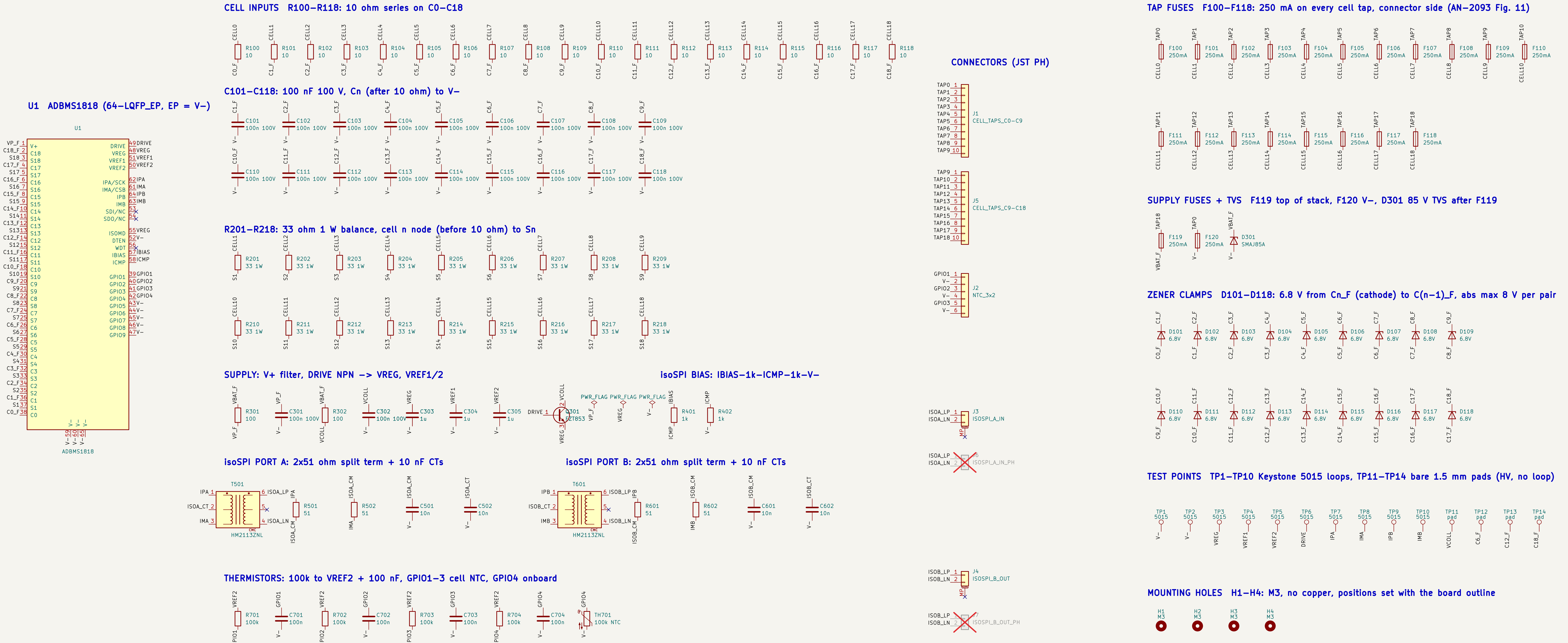
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.



U1 ADBMS1818 (64−LQFP_EP, EP = V−)

CELL INPUTS R100−R118: 10 ohm series on C0−C18

C101−C118: 100 nF 100 V, Cn (after 10 ohm) to V−

R201−R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn

SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2

isoSPI BIAS: IBIAS−1k−ICMP−1k−V−

isoSPI PORT A: 2x51 ohm split term + 10 nF CTs

isoSPI PORT B: 2x51 ohm split term + 10 nF CTs

THERMISTORS: 100k to VREF2 + 100 nF, GPIO1−3 cell NTC, GPIO4 onboard

CONNECTORS (JST PH)

TAP FUSES F100−F118: 250 mA on every cell tap, connector side (AN−2093 Fig. 11)

SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119

ZENER CLAMPS D101−D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair

TEST POINTS TP1−TP10 Keystone 5015 loops, TP11−TP14 bare 1.5 mm pads (HV, no loop)

MOUNTING HOLES H1−H4: M3, no copper, positions set with the board outline

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User Date: 2026−11−06 **Rev: 0−draft**
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